

CHAPTER – 9

JAVA FULLSTACK SUMMARY

- **What is Exception and its types .**
- **Multiple catch blocks .**
- **Finally block .**
- **Difference between throw and throws.**
- **Custom Exception.**
- **Lambda Function.**

WHAT IS EXCEPTION AND ITS TYPES :-

- Exception is a abnormal event that occurs during the execution of a program.
- Disrupts the normal flow of instructions.
- Exceptions can be handled so that the program does not terminate unexpectedly.

TYPES OF EXCEPTION:-

- ✓ Checked exception.
- ✓ Unchecked exception.
- ✓ Errors.

CHECKED EXCEPTION:-

- These are exceptions checked and enforced by the compiler.
- You must to handle them using try-catch and throws.

EXAMPLES:-

- **IOException**
- **SQLException**
- **ClassNotFoundException.**
- **FileNotFoundException**

UNCHECKED EXCEPTION:-

- These are occurs during the runtime and are not checked by the compiler.

EXAMPLE:-

- ArithmeticException.
- NullPointerException.
- ArrayIndexOutOfBoundsException.
- NumberFormatException.

ERRORS:-

- Errors are serious issues that are not meant to be handled by the applications.

EXAMPLE:-

- OutOfMemoryError
- StackOverflowError
- VirtualMachineError

MULTIPLE CATCH BLOCKS:-

- ✚ A multiple catch block means using more than one catch block with a single try block to handle different types of exceptions separately.
- ✚ A single try block they can have multiple try blocks.
- ✚ The most specific exceptions should come before the more general exception.
- ✚ The exception class should usually be the last catch block.

FINALLY BLOCK:-

- ✚ A finally block is used to execute code whether an exception occurs or not
- ✚ Its mainly used for cleanup operations such as closing files, database connections or releasing resources.
- ✚ The finally block always execute, whether an exception occurs or not.
- ✚ It is optional and used for cleanup the code.

DIFFERENCE BETWEEN THROW AND THROWS:-

Throw	Throws
Used to explicit throw an exception	Used to declare exception that a method may throw
Used inside a method	Used in the method signature
Throws one exception at a time	Can declare multiple exceptions

CUSTOM EXCEPTION:-

- ✚ A custom exception is a user defined exception created by the programmer to handle the applications -m specific errors.
- ✚ It is created by extending the exception class or Runtime Exception class.
- ✚ To handle applications specific errors.
- ✚ Makes the code more readable.

LAMBDA FUNCTION:-

- ✚ A lambda expression is a short way to write an anonymous function
- ✚ It was introduced in java 8 to make code shorter and easier to read.